Quadratic & Cubic Sequences

Exam code: 0607

Download PDF

Written by: Mark Curtis
Reviewed by: Dan Finlay
Updated on 27 November 2025

Quadratic & Cubic Sequences

What is a quadratic sequence?

- A **quadratic** sequence has an ***n* th term formula** that involves n^2
- The **second differences** are **constant** (the same)
  - These are the differences between the first differences
  - For example, 3, 9, 19, 33, 51, ...
    1st Differences: 6, 10, 14, 18, ...
    2nd Differences: 4, 4, 4, ...

How do I find the nth term formula for a simple quadratic sequence?

- The sequence with n th term formula n^2 are the **square numbers**
  - 1, 4, 9, 16, 25, 36, 49, ...
    - From $1^2, 2^2, 3^2, 4^2, \ldots$
- Finding the ***n* th term formula** comes from **comparing** sequences to the square numbers
  - 2, 5, 10, 17, 26, 37, 50, ... has the formula $n^2 + 1$
    - Each term is 1 more than the square numbers
  - 2, 8, 18, 32, 50, 72, 98, ... has the formula $2n^2$
    - Each term is double a square number
- It may be a simple **combination** of both
  - For example, doubling then adding 1
    - 3, 9, 19, 33, 51, 73, 99, ... has the formula $2n^2 + 1$
- Some sequences are just the square numbers but starting later
  - 16, 25, 36, 49, ... has the formula $(n + 3)^2$
    - Substitute in $n = 1$, $n = 2$, $n = 3$ to see why

What is a cubic sequence?

- A **cubic** sequence has an ***n* th term formula** that involves n^3
- The **third differences** are **constant** (the same)
  - These are the differences between the second differences

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

3rd Differences: 18, 18, 18, ...

How do I find the nth term formula for a simple cubic sequence?

- The sequence with n th term formula $\boldsymbol{n^3}$ are the **cube numbers**
  - 1, 8, 27, 64, 125, ...
    - From $1^3, 2^3, 3^3, 4^3, \ldots$
- Finding the ***n* th term formula** comes from **comparing** sequences to the cube numbers
  - 2, 9, 28, 65, 126, ... has the formula $n^3 + 1$
    - Each term is 1 more than the cube numbers
  - 2, 16, 54, 128, 250, ... has the formula $2n^3$
    - Each term is double a cube number

Can I use the differences to help find the nth term formula?

- Yes, for the simple **quadratic** sequence $an^2 + b$
  - a is $\frac{1}{2}$ of the **second** difference
- For the simple **cubic** sequence $an^3 + b$
  - a is $\frac{1}{6}$ of the **third** difference

Examiner Tips and Tricks

- You must learn the square numbers from 1^2 to 15^2 and the cube numbers from 1^3 to 5^3 (and know 10^3)

Worked Example

For the sequence 6, 9, 14, 21, 30, ...

(a) Find a formula for the nth term.

Answer:

Find the first and second differences

Sequence: 6, 9, 14, 21, 30
First differences: 3, 5, 7, 9, ...
Second differences: 2, 2, 2, ...

The second differences are constant so this must be a quadratic sequence
Compare each term to terms in the sequence n^2 (the square numbers)

Each term is 5 more than the terms in n^2, so add 5 to n^2

n^{th} term = $n^2 + 5$

You could also have used $an^2 + b$ where a is half the second difference

(b) Hence, find the 20th term of the sequence.

Answer:

Substitute $n = 20$ into $n^2 + 5$

$$(20)^2 + 5 = 400 + 5$$

The 20th term is 405

Teacher only

Test your students on this in just a few clicks

- Create exam-aligned tests
- Tailor format & difficulty
- Quizzes or homework in minutes

Try Test Builder →

Test yourself

Was this revision note helpful? Yes No

Previous Next

Course Study tools All topics

Resources

Learning Hub

Scholarship

Join

Past Papers

Dashboard

Overview

Brand ambassador

Profile

Subscription

Company

About us

Exam Specificity

Content Quality

Promotions

Jobs

Terms

Privacy

Cookie Policy

Help and Support

Subjects

Biology

Chemistry

Physics

Maths

Geography

English Literature

Psychology

All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.

Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.